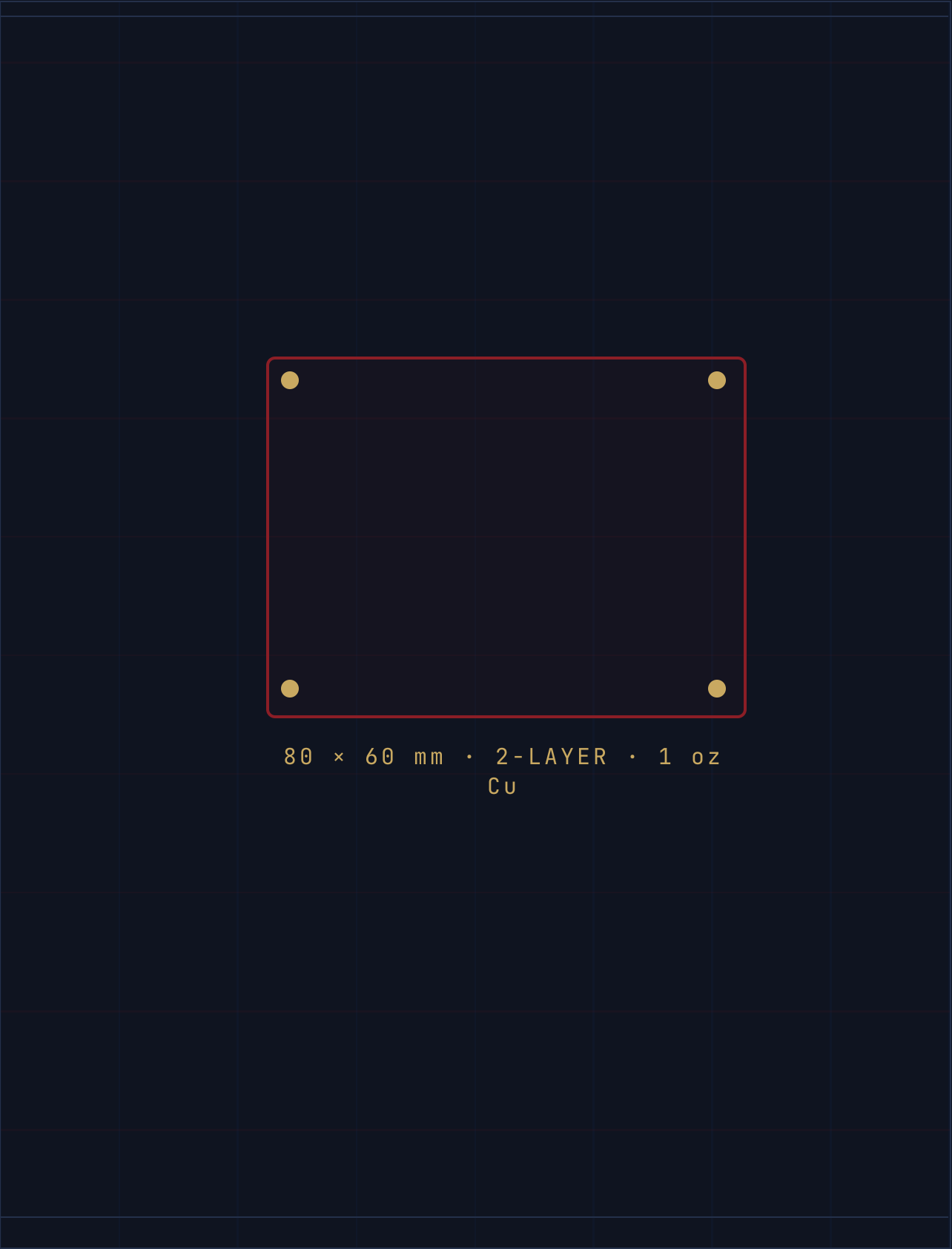


Project

Tight-Fit

Main-Control PCB · *Team Briefing*

3 HOURS · 4 ENGINEERS · 80 × 60 mm · PROJECTED 100 / 100



Consolidate 8 modules onto *one* 80 × 60 mm board.

Rover ships with spaghetti wiring — the challenge is a single, ruled-up main-control PCB.

THE 8 MODULES • 1 BOARD

- 01 ESP32 DevKit V1 — MCU
- 02 L298N — dual H-bridge motor driver
- 03 NEO-6M — GPS receiver (UART)
- 04 HC-SR04 — ultrasonic (5 V logic)
- 05 BSS138 × 4 — bidirectional level shifter
- 06 MP1584 — buck converter → 5 V
- 07 XT60 · 2 A fuse · SPST 3 A switch
- 08 3S LiPo — 11.1 V nom / 12.6 V full

RUBRIC • 100 PTS

SSch	Schematic quality	30
SPCB	PCB layout	40
SEng	Engineering rules	20
SDoc	Documentation	10

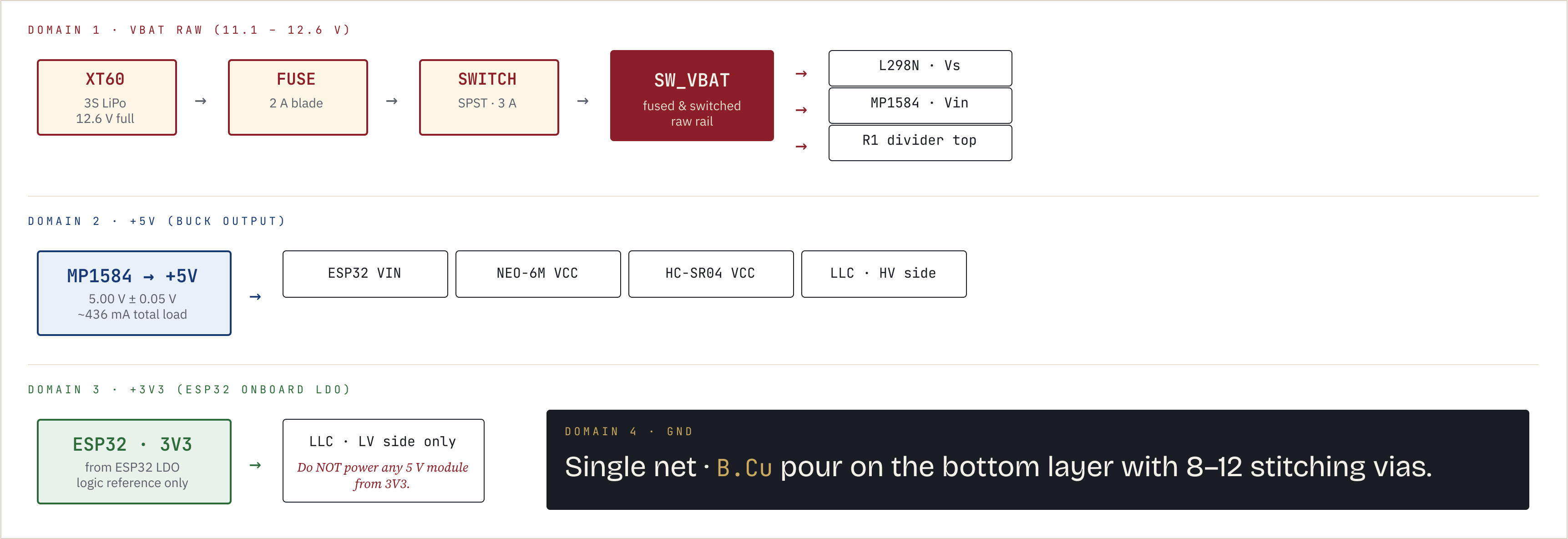
HARD RULE

AI PCB generators *banned*.

Penalty • -100 pts

Four power domains. *One* ground pour.

Battery → fused/switched raw → 5 V rail → 3.3 V rail. Route power first, signals second.



The fuse is the *weakest* link — by design.

Ratings ordered so the fuse always blows before the switch welds and before the copper anneals.

SAFETY INEQUALITY

$$2.0\text{ A} \leq 3.0\text{ A} \leq 2.4\text{ A}$$

FUSE · SWITCH · TRACE (continuous)

FUSE · 2.0 A

First to sacrifice.

Sized below every downstream element. Opens the circuit on a short before energy reaches the switch or the copper. Cheap, replaceable, deterministic.

SWITCH · 3.0 A

Never sees fault current.

50 % headroom above fuse rating means the SPST contacts never carry a fault. Prevents contact welding — a switch that welds is stuck ON.

TRACE · 2.4 A @ ΔT 10 °C

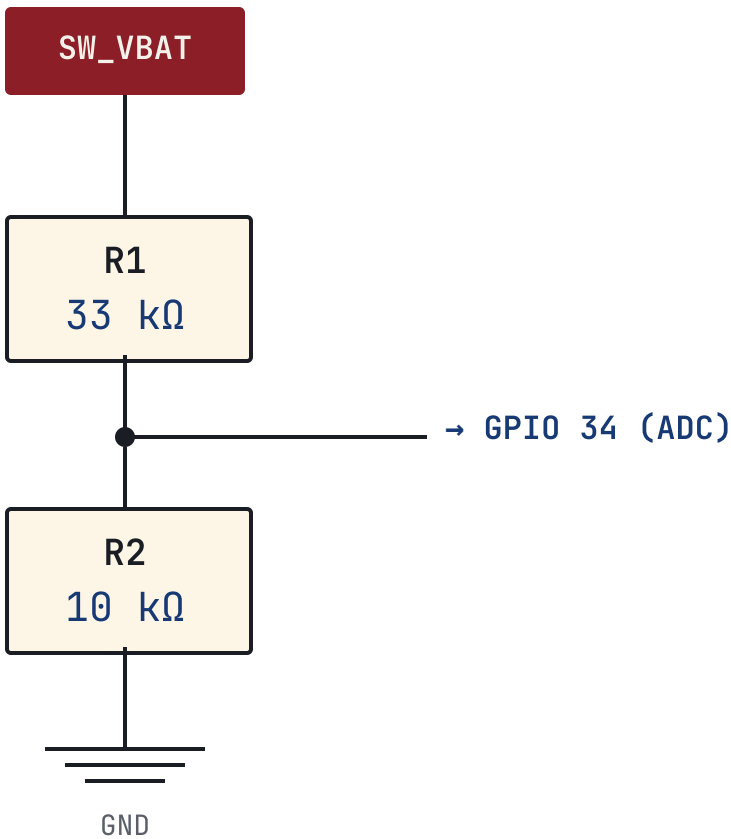
Copper never anneals.

30 mil trace (see §07) carries 2.4 A continuous with a 10 °C rise. Even a sustained overload stays inside copper's thermal envelope until the fuse opens.

R1 = 33 kΩ · R2 = 10 kΩ.

Both 0805, 1 % tolerance. Scales the 12.6 V full-battery down to a safe ADC input on GPIO 34.

DIVIDER · SCHEMATIC



Both resistors 0805 SMD, 1 % tolerance. Placed adjacent to ESP32 to keep the ADC trace short.

DIVIDER MATH · FULL CHARGE

$$\begin{aligned} V_{\text{out}} &= V_{\text{in}} \times R2 / (R1 + R2) \\ &= 12.6 \times 10 / 43 \\ &= 2.930 \text{ V} \end{aligned}$$

ADC HARD LIMIT

3.30
V

ESP32 ADC absolute max (VDD 3.3 V rail).

DIVIDER OUTPUT @ 12.6 V

2.930
V

Nominal ADC reading at full-charge battery.

HEADROOM

0.37
V

Below the 3.3 V ADC ceiling — safe margin.

Both constraints hold — *including* 1 % worst case.

Safety bound = must not exceed 3.0 V. Usability bound = must exceed 2.5 V for readable dynamic range.

CONSTRAINT A • SAFETY

$V_{\text{out}} \leq 3.0 \text{ V}$ — never damages the ADC pin.

$$2.930 \text{ V} \leq 3.000 \text{ V}$$

Nominal reading sits **70 mV** under the hard limit — pass. ✓

CONSTRAINT B • USABILITY

$V_{\text{out}} \geq 2.5 \text{ V}$ — keeps useful ADC range.

$$2.930 \text{ V} \geq 2.500 \text{ V}$$

Cell-empty warning threshold well resolved — pass. ✓

1 % WORST-CASE SWEEP

$$2.885 - 2.975 \text{ V}$$

Both bounds still hold across the full tolerance band. Max 2.975 V < 3.0 V. Min 2.885 V > 2.5 V.

QUIESCENT CURRENT FROM DIVIDER

$$I_q = 293 \text{ } \mu\text{A}$$

Divider alone would flatten a 3S LiPo in **~312 days** — negligible next to motor loads.

30 mil on all four power nets — $\Delta T 10.9\text{ }^{\circ}\text{C @ }2\text{ A}$.

External-layer, 1 oz Cu. Derived from IPC-2221 for continuous 2 A carry.

IPC-2221 • EXTERNAL TRACE, 1 OZ CU

$$I = k \cdot \Delta T^{0.44} \cdot A^{0.725}$$

k 0.048 (external layer)

ΔT 10 °C target rise

H 1.378 mil (1 oz Cu)

A W × H (cross-section, mil²)

Solve for I = 2 A:

$$A \geq 39.3 \text{ mil}^2$$

$$W \geq A / H = 39.3 / 1.378$$

$$W \geq 28.5 \text{ mil}$$

CHOSEN WIDTH

30 mil

$$= 0.762 \text{ mm}$$

Actual ΔT at 2 A continuous · 10.9 °C

APPLIED TO • PWR NET CLASS

VBAT_RAW • VBAT_FUSED • SW_VBAT • +5V

Signal nets (I/O, UART, ADC) stay on the Default class at 0.25 mm — different rule, same board. Set both in KiCad's net-class table before routing.

Every pin has *one* role. Strap pins stay clear.

Boot-strap pins 0, 2, 5, 12, 15 avoided for any critical output. ADC on ADC1 only (Wi-Fi safe).

SUBSYSTEM	SIGNAL	GPIO	FUNCTION	NOTE
L298N	IN1 / IN2 / IN3 / IN4	25 · 26 · 27 · 14	Digital out	Motor direction bits
L298N	ENA / ENB	32 · 33	PWM out	Speed control
NEO-6M	UART2 RX / TX	16 · 17	Hardware UART	3.3 V native — direct
HC-SR04	TRIG → LLC CH1	13	3.3 → 5 V out	Via level shifter
HC-SR04	ECHO ← LLC CH2	18	5 → 3.3 V in · MANDATORY	Direct 5 V damages pin
Telemetry	VBAT sense	34	ADC1 in	Wi-Fi safe · input-only

STRAP · **avoided** · GPIO 0, 2, 5, 12, 15 — reserved for boot mode / flashing. Never assigned to a critical output.

ECHO is *worth 8 points* — do not skip.

Two channels used, two unpopulated. GPS channels stay unpopulated because standard NEO-6M breakouts are 3.3 V native.

CH1 · USED TRIG 3.3 V → 5 V SOURCE · SINK ESP32 GPIO 13 → HC-SR04 TRIG pin WHY SHIFT? HC-SR04 wants a clean 5 V trigger. A 3.3 V CMOS-high is right on the module's threshold — the shift removes the ambiguity. POPULATE	CH2 · MANDATORY ECHO 5 V → 3.3 V SOURCE · SINK HC-SR04 ECHO → ESP32 GPIO 18 SKIP COST Direct 5 V into a 3.3 V GPIO exceeds the abs-max input and can permanently damage the pin. Rubric line item worth 8 points. POPULATE	CH3 · CONDITIONAL GPS · RX only if 5 V logic THIS BUILD Standard NEO-6M breakout — 3.3 V native (verified via on-board LDO). VERDICT Footprint on the schematic and PCB but no shifter part fitted. Direct UART wire is correct here. UNPOPULATED	CH4 · CONDITIONAL GPS · TX only if 5 V logic THIS BUILD Same reasoning as CH3 — NEO-6M breakout is 3.3 V. VERDICT Reserved on the board. Fit the shifter part only if a future 5 V-logic GPS module is used. UNPOPULATED
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MP1584 over LM2596 — *and* tune it before you mount it.

Same task, smaller footprint, higher efficiency, lower ripple — but the pot ships wrong.

WHY MP1584		
METRIC	MP1584 ✓	LM2596
Footprint	22 × 17 mm	43 × 20 mm
Efficiency	92 %	~ 78 %
Ripple	low	higher
Rating	3 A	3 A

5 V LOAD HEADROOM

6.9×

~436 mA total load • 3 A rating

ESP32 • NEO-6M • HC-SR04 • LLC HV combined.

WARNING • PRE-MOUNT POT PROCEDURE

Modules ship with the trim-pot set to > 12 V. Mounting a mis-tuned MP1584 fries every 5 V module.

6-step tune • bench PSU 12 V in → DMM on Vout → turn pot CCW → read → repeat → target 5.00 V ± 0.05 V • then mount to PCB.

Steps 1–11 · *ERC = 0* before you touch the PCB editor.

Order matters. Power domains first, signals second — same rule as the PCB.

STEPS 1–6 · STRUCTURE

- 01 Project setup.** New KiCad project; add ESP32-DevKitV1, L298N, NEO-6M, HC-SR04, MP1584, BSS138 × 4, XT60, fuse, switch symbols.
- 02 Visual grouping.** Draw labelled rectangles per power domain — VBAT, +5V, +3V3. Readable schematic = SDoc points.
- 03 VBAT chain.** XT60 → 2 A fuse → SPST switch → SW_VBAT rail. This order — fuse BEFORE switch — is scored (§14).
- 04 5 V / 3V3 distribution.** Buck out feeds ESP32 VIN, NEO-6M, HC-SR04, LLC HV. ESP32 3V3 feeds LLC LV *only*.
- 05 Divider.** R1 33 kΩ, R2 10 kΩ (0805 · 1 %) from SW_VBAT to GND, tap → GPIO 34.
- 06 Motor control.** L298N IN1..IN4 = GPIO 25/26/27/14, ENA/ENB = GPIO 32/33 (PWM).

STEPS 7–11 · FINISH

- 07 GPS UART.** NEO-6M RX/TX ↔ ESP32 GPIO 17/16. Direct wire, no shifter (breakout is 3.3 V native).
- 08 Ultrasonic thru LLC.** TRIG (GPIO 13) → LLC CH1 → HC-SR04. ECHO ← LLC CH2 ← HC-SR04 → GPIO 18. **ECHO mandatory.**
- 09 Annotate.** Tools → Annotate Schematic — every reference designator resolved, no ? left.
- 10 ERC.** Target **0 errors**. Add PWR_FLAG on VBAT, +5V, +3V3 to silence phantom power-net warnings.
- 11 Footprints.** Assign SMD only — 0805 / 1206 for R, C. Any THT R/C zeroes SEng (see §14).

Steps 12–18 · *DRC = 0*. Route power, then signals.

Fixed placement, fixed order — the layout choices are pre-solved so you can execute at speed.

STEPS 12–14 · SETUP

- 12 **Open PCB editor** · pull netlist from schematic.
- 13 **Net classes.**
PWR → 0.762 mm (30 mil)
Default → 0.250 mm
- 14 **Board outline.**
Edge.Cuts → 80 × 60 mm rectangle
4 × M3 (Ø 3.0 mm) holes at:
 (3, 3) (77, 3)
 (3, 57) (77, 57)

STEPS 15–18 · PLACE · ROUTE · POUR · VERIFY

- 15 **Placement order.**
XT60 → Fuse → Switch → Buck
 → L298N → ESP32 (central)
 → LLC → GPS + HC-SR04 (opposite edge)
- 16 **Route power first, signals second.** Fat power on F.Cu, signals threaded around.
 Never route power through the pour.
- 17 **Ground pour.** Full B.Cu GND fill · 8–12 stitching vias to knit F.Cu GND islands.
- 18 **DRC.** Target **0 errors**. Refill zones and re-DRC after every routing change.

Five labels · *five* free points.

Silkscreen is the last SDoc line item — cheap to add, and the last place we let points bleed out.

THE 5 LABELS · F.SILKSCREEN

01	+ / -	polarity at XT60 connector	+1 pt
02	3.3V	at LLC LV side pin row	+1 pt
03	5V	at MP1584 buck output	+1 pt
04	VBAT TP	at test-point pad	+1 pt
05	ON ► / ◀ OFF	at SPST switch	+1 pt

*Nets total declared in title block · **N_total** = 19. Add to project properties before export.*

COMMON ERC / DRC PITFALLS

Expect these — check for them.

- PITFALL 01

Unrouted nets

Toggle ratsnest layer · zero remaining airwires.
- PITFALL 02

Clearance under ESP32 pads

Drop a via-in-pad reminder — recheck castellation clearance.
- PITFALL 03

Unfilled pour

Re-fill zones (B) before every DRC — KiCad does not auto-refill.
- PITFALL 04

Pour islands

Add a stitching via at the island to re-join it to GND.

Predicted *100 / 100*. Do not improvise.

SSCH · SCHEMATIC

30

ERC = 0 · PWR_FLAG · grouped

SPCB · LAYOUT

40

DRC = 0 · 30 mil PWR · pour

SENG · RULES

20

SMD only · protection order · LLC

SDOC · DELIVERABLES

10

5 silks · math sheet · pinout

TOTAL (PREDICTED)

100 / 100

TEAM TRAPS · WHAT ERASES THE SCORE

ECHO wired direct to ESP32 (no LLC)

- 8

Any THT R / C on the board

- 20 (SEng zeroed)

Fuse placed AFTER switch

- 3

Missing / unfilled ground pour

- 5

DELIVERABLES · ONE ZIP

└ schematic.pdf

└ pcb.pdf

└ math_sheet.pdf

└ pinout.pdf

Execute in KiCad. Do not improvise the power path.